

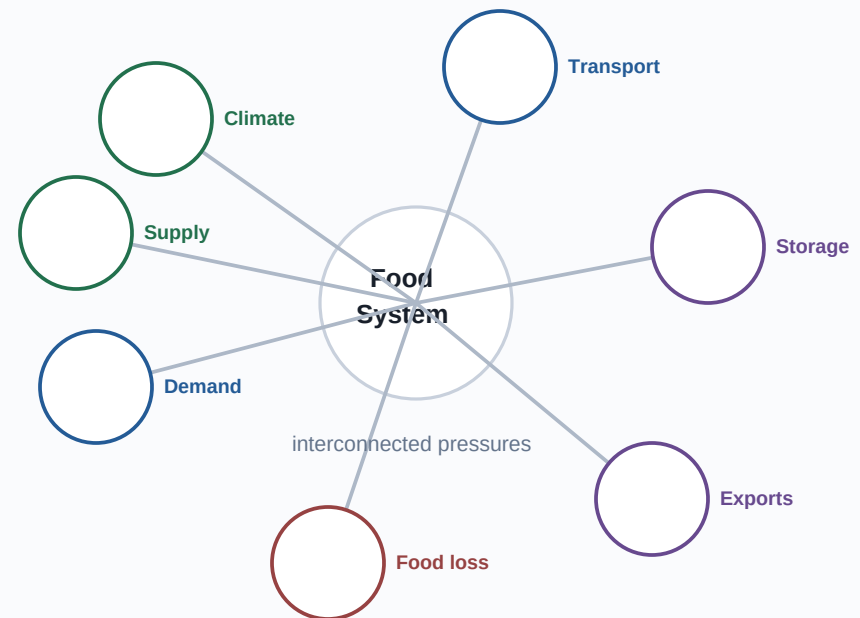
Systems Under Pressure

The Mathematics of Food Systems

"All models are wrong, but some are useful."

George Box

A concise investigation brief for studying real food systems under mathematical pressure.





"All models are wrong, but some are useful."

George Box

This project is not about perfect certainty. It is about using mathematics to make a complex system clearer, then naming what the mathematics still cannot see.

PROJECT PHILOSOPHY

A model is useful, but incomplete.

You are not simply completing Algebra 2 tasks. You are investigating an unstable food system and using mathematics to make its behavior more visible. Models reveal patterns, hide variables, depend on assumptions, and eventually break down. Strong mathematical thinkers name those limits honestly.

**Revision is evidence of
mathematical thinking.**

Think like a mathematical investigator.

This project values interpretation over memorization and visible reasoning over polished certainty.

Models reveal

patterns, rates,
structure, constraints,
thresholds, and possible
futures

Models hide

unmeasured variables,
human decisions, sudden
disruptions, uncertainty,
and local context

Models change

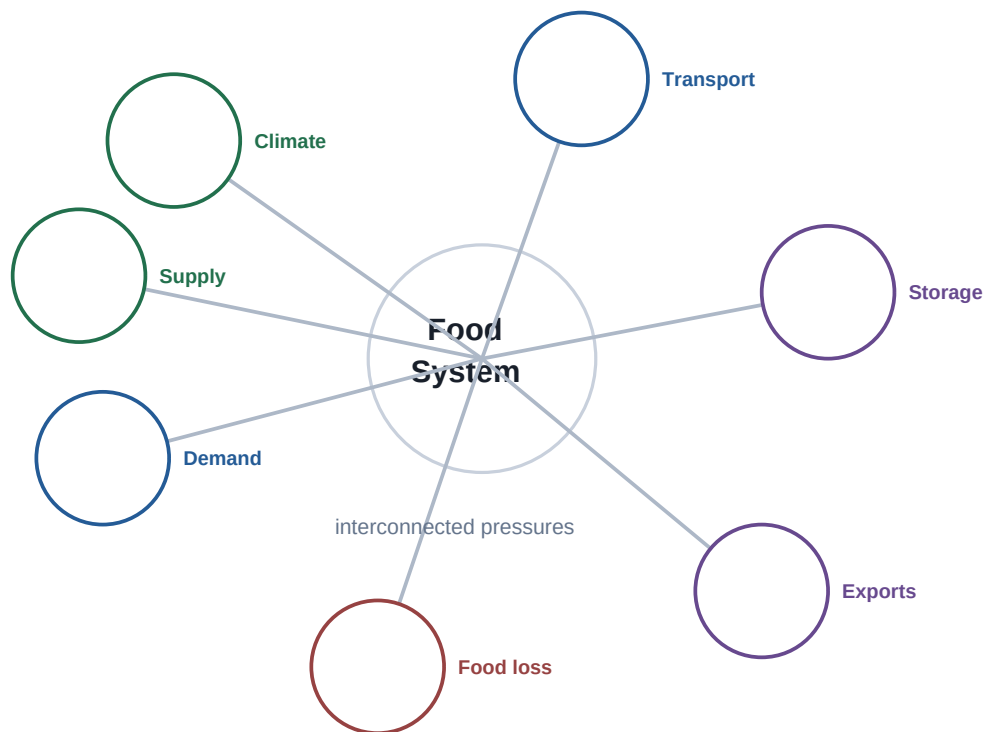
when assumptions, rates,
dimensions, domains, or
constraints are revised

WHAT COUNTS AS STRONG THINKING

Crossed-out assumptions, recalculated thresholds, revised graphs, changed interpretations, and honest limitations are not mistakes to hide. They are evidence that the investigation became more sophisticated.

One system. Many pressures.

Food systems are interconnected. Changes in one part can reshape the whole system.



Mathematics helps us

Model change, identify patterns, compare possibilities, estimate outcomes, critique predictions, and reveal constraints.

Strong thinkers ask

What assumptions are we making? What variables matter most? Where does the model stop making sense? What tradeoffs appear?

Choose the format that best communicates the investigation

The purpose is communicating mathematical reasoning, not producing a polished slideshow.

Allowed formats

slides, filmed
discussions, workbook
walkthroughs, digital
whiteboards, screencasts,
annotated journals,
documentary-style
explanations, or hybrid
combinations

Math remains central

models, graph analysis,
comparisons, assumptions,
revisions, limitations,
final claims, and visible
mathematical evidence

Clarity over polish

Do not prioritize
cinematic editing,
effects, overproduction,
or memorized performance.
Prioritize reasoning and
explanation.

WORKBOOK WALKTHROUGHS ARE ENCOURAGED

The journal is a major part of the investigation. Students may explain crossed-out assumptions, revised graphs, recalculations, and changing interpretations directly from the workbook.

What makes the investigation visible

These pieces work together. The journal shows the evolving inquiry; the final communication explains the mathematical story.

Investigation Journal

rough work, sketches, Desmos screenshots, calculations, AI interactions, revisions, and evolving reasoning

Mathematical Artifacts

graph annotations, recursive calculations, polynomial long division, tables, recalculations, and revised models

Group Reflection Vlogs

short mathematical conversations showing uncertainty, revision, disagreement, critique, and progress

Final Communication

the chosen format that communicates models, comparisons, assumptions, limitations, and claims

One shared investigation. Visible individual thinking.

Do not divide units by student. The group functions as a collaborative mathematical inquiry team.

COLLABORATIVE WORK

Students think together, discuss together, graph together, revise together, compare together, critique together, and synthesize together.

INDIVIDUAL EVIDENCE SNAPSOTS

Each student contributes smaller, frequent artifacts: graph interpretations, model comparisons, recalculations, AI critique, revised assumptions, or limitation reflections.

EVERY STUDENT SPEAKS MATHEMATICS

Every student must visibly contribute mathematical reasoning, interpretation, comparison, critique, or explanation in group vlogs and final communication. The goal is authentic reasoning, not mechanical speaking turns.

Choose one documentary investigation prompt.

Each scenario begins with a human tension. Your group narrows it into one measurable mathematical investigation.

A | United States

The \$5 Footlong Isn't \$5 Anymore

Changing everyday food costs reshape affordability.

Key tensions: inflation, pricing, shrinkflation, labor, transportation

B | Global

When War Changes the Price of Bread

Disruption in one region can shift prices and recovery patterns elsewhere.

Key tensions: exports, wheat, transport, recovery, supply chains

C | Sub-Saharan Africa

What If Food Exists - But Never Arrives?

Food can be produced and still be lost before reaching people.

Key tensions: spoilage, refrigeration, storage, infrastructure

D | Colombia

Can Climate Change the Future of Coffee?

Environmental and economic changes can alter yield, exports, and thresholds.

Key tensions: rainfall, coffee yield, climate, transport, exports

A good question is narrow enough to model.

Avoid giant topics. Look for one changing quantity, one measurable relationship, and one meaningful constraint.

Too broad

How inflation affects food
Climate and coffee
Food insecurity
Food waste

Better

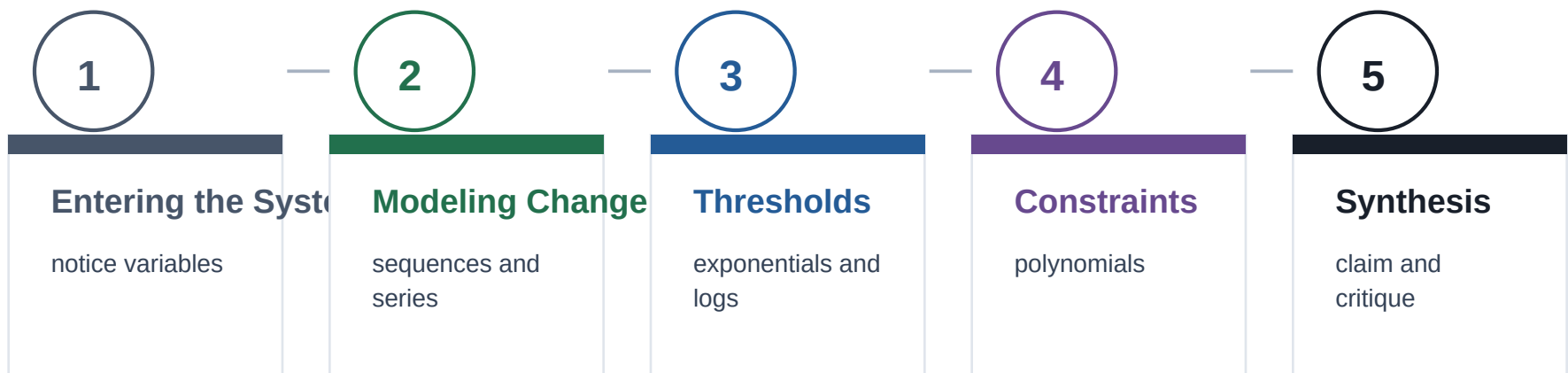
How sandwich prices changed over time
Rainfall vs coffee yield in one region
Food spoilage during transport stages
Food remaining after repeated losses

STRONG FOCUS TEST

Can your group identify one main changing quantity, one measurable relationship, one prediction or threshold question, one physical or cost constraint, and one connection across mathematical lenses?

The five-class arc is one evolving investigation.

Each phase builds on the same system. The mathematics becomes more sophisticated through comparison and revision.



PROJECT ARC

Notice relationships, model repeated change, recognize acceleration and thresholds, model constraints, then synthesize what each model reveals and misses.

Different structures reveal different things.

Students repeatedly answer: Why this model? What does it reveal? What does it miss? How do I know?

Sequences & Series

How is the system changing over time? Compare discrete vs continuous, arithmetic vs geometric, recursive vs explicit, finite vs infinite.

Exponentials & Logs

When does change accelerate, decay, compound, or cross a threshold? Compare linear vs exponential and sampled values vs continuous models.

Polynomials

How do capacity, cost, and physical constraints reshape the system? Analyze structure, end behavior, extrema, and domain.

LIGHTWEIGHT MODEL FIT

Use r and R^2 as credibility and critique tools. Interpret direction, strength, fit, hidden variables, and causation limits. This is not a full statistics unit.

Three content standards. Three skill grades.

The final rubric separates Algebra 2 content from cross-cutting mathematical habits. Both matter.

CORE ALGEBRA 2 CONTENT STANDARDS

1. Sequences and Series

discrete change,
arithmetic vs geometric
patterns, recursive vs
explicit forms, finite vs
infinite reasoning,
cumulative sums in
context

2. Exponentials and Logarithms

growth and decay, growth
factors, thresholds,
logarithmic reasoning,
realistic interpretation
and limitations

3. Polynomials

degree, leading
coefficient, end
behavior, intercepts,
multiplicity, extrema,
polynomial forms, and
realistic domain
restrictions

CROSS-CUTTING SKILL GRADES

4. Mathematical Communication

clear explanations,
graph-algebra-context
connections,
assumptions,
annotations, tables, and
contextual
interpretation

5. Comparing Models, Assumptions

model comparison,
realism, hidden
variables, tradeoffs,
assumptions,
unreasonable
predictions, and
meaningful revision

6. Algebraic Precision and Flexible N

accurate methods,
factoring, logarithms,
substitution, graph
interpretation, rational
numbers, units, and
reasonableness checks

SAIL L5: Co-Create

AI is a thinking partner, not an answer machine. It should deepen reasoning and make critique sharper.

AI may help you

challenge assumptions, critique models, identify missing variables, compare interpretations, explain r and R^2 , organize ideas, and revise explanations

AI may not replace you

Do not submit reasoning you do not understand, ask AI to complete the project, hide meaningful AI use, or replace your own mathematical voice.

TRANSPARENCY SENTENCE

AI use: We asked ChatGPT to critique our constant-growth assumption. We rejected one suggestion, revised our rate assumption, and recalculated the threshold.